AUTOSAR High-Level Design

Synthetic Demonstration Document · Version 1.0

Purpose: Demonstrate document analysis, entity extraction, evidence retrieval, and revision comparison in the AUTOSAR HLD Document Analysis Assistant.

Important: This sample is fictional and intended only for software demonstration. It is not a production vehicle architecture or a compliance artifact.

Document control

Document ID: DEMO-HLD-001
Project: Demo Vehicle Control Platform
Owner: Systems Engineering
Status: Draft for review
Classification: Synthetic sample

1. System Overview

The Demo Vehicle Control Platform illustrates a simplified AUTOSAR-style architecture. The system contains VehicleStateProvider, BrakeManager, PowertrainCoordinator, DiagnosticGateway, and CabinDisplayManager. Components communicate through named ports and signals. All entities in this sample are illustrative.

VehicleStateProvider publishes VehicleSpeedSignal and BrakePedalSignal. BrakeManager consumes VehicleSpeedSignal and BrakePedalSignal to calculate a candidate braking request. PowertrainCoordinator consumes VehicleSpeedSignal and publishes EngineTorqueRequestSignal. CabinDisplayManager consumes VehicleSpeedSignal to display current vehicle speed.

2. Software Components

Component	Responsibility	Inputs	Outputs
VehicleStateProvider	Aggregates vehicle-state data from approved sensors	SensorDataPort	VehicleSpeedSignal, BrakePedalSignal
BrakeManager	Computes candidate braking request from vehicle-state inputs	VehicleSpeedPort, BrakePedalPort	BrakeRequestSignal
PowertrainCoordinator	Coordinates torque request using vehicle speed and driver demand	VehicleSpeedPort	EngineTorqueRequestSignal
DiagnosticGateway	Routes diagnostic requests to component diagnostic handlers	DiagnosticRequestPort	DiagnosticResponseSignal
CabinDisplayManager	Displays speed and selected vehicle status	DisplayInputPort	DisplayStatusSignal

3. Interfaces and Ports

VehicleSpeedPort is the receiver-side port used by BrakeManager, PowertrainCoordinator, and CabinDisplayManager. SensorDataPort is the input port for VehicleStateProvider. BrakePedalPort provides BrakePedalSignal to BrakeManager. DiagnosticRequestPort and DiagnosticResponseSignal support the diagnostic path. DisplayInputPort receives selected vehicle-state values for the cabin display.

4. Signals and Data Flow

VehicleStateProvider publishes VehicleSpeedSignal. BrakeManager consumes VehicleSpeedSignal and BrakePedalSignal and produces BrakeRequestSignal. PowertrainCoordinator consumes VehicleSpeedSignal and produces EngineTorqueRequestSignal. CabinDisplayManager consumes VehicleSpeedSignal and produces DisplayStatusSignal.

5. Dependencies

BrakeManager depends on VehicleStateProvider for VehicleSpeedSignal and BrakePedalSignal. PowertrainCoordinator depends on VehicleStateProvider for VehicleSpeedSignal. CabinDisplayManager depends on VehicleStateProvider for VehicleSpeedSignal. DiagnosticGateway uses DiagnosticRequestPort and returns DiagnosticResponseSignal.

6. Constraints and Assumptions

All communication periods, timing budgets, safety classifications, and AUTOSAR release-specific configuration details are intentionally omitted from this synthetic sample. They must be sourced from approved project requirements and validated by responsible engineers.

7. Review Notes

The architecture names are illustrative. The analysis application should preserve source-page citations, identify uncertainty, and never automatically approve architecture decisions. A lack of automated flags does not imply that the design is correct or complete.

8. Traceability Reference

REQ-DEMO-001: Vehicle speed shall be made available to consumer components through an approved interface.
REQ-DEMO-002: BrakeManager shall consume the required vehicle-state signals before producing a braking request.
REQ-DEMO-003: Diagnostic request and response paths shall be documented and reviewed.
TEST-DEMO-001: Verify VehicleSpeedSignal is available to each listed consumer.
TEST-DEMO-002: Verify diagnostic request routing in an approved test environment.

9. Revision History

Revision	Date	Change
1.0	2026-09-01	Initial synthetic demo architecture

End of synthetic demonstration document.